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ELEN VARDANYAN

linkedin.com/in/elenvardanyan

WORK EXPERIENCE

Adjunct Lecturer
AMERICAN UNIVERSITY OF ARMENIA (AUA)
AUGUST 2022 - 2024

- Conducted lectures for "*Machine Learning*" and "*Deep Learning*" courses for bachelor's and master's students in different programs
- Prepared and gave feedback on assignments, quizzes and exams
- Supervised capstone projects and theses in the topics:
  - "*Vehicle License Plate Recognition in Armenia*"
  - "*Real-Time Parking Occupancy Detection using Deep Learning*"
  - "*Armenian speech-to-text recognition*"
  - "*Intelligent university chatbot*", etc.

Junior Researcher
YEREVAN STATE UNIVERSITY (YSU), FAST Foundation
APRIL 2023 - AUGUST 2024

- Researcher in the group "*Statistical Analysis of Machine Learning Algorithms (SAM-lab)*", FAST ADVANCE program
- Topic: "*Statistical Guarantees of Generative Models*" supervised by *Professor Arnak Dalalyan*

Machine Learning Researcher
YEREVAN STATE UNIVERSITY (YSU) AI LAB
APRIL 2023 - APRIL 2024

- Collaborated on industry problems proposed by partner companies such as *PMI Science* and *ServiceTitan*
- Worked on finetuning and prompt engineering for LLMs
- Led small groups of mid and junior researchers

Adjunct Lecturer
YEREVAN STATE UNIVERSITY (YSU)
OCTOBER 2022 - JUNE 2024

- Conducted lectures for "*Introduction to Data Science*" and "Machine Learning 2" courses for bachelor's and master's students
- Prepared and gave feedback on assignments, quizzes and exams

Data Scientist
PICSART, INC.
SEPTEMBER 2017 - JULY 2019

- Performed ETL, EDA as well as production-ready model development for *customer churn prediction, recommendation systems,* and *hashtag trend detection* in the User Behavior Group
- Worked on *image inpainting* and *sticker recommendation on pictures* in distributed environments in the Computer Vision Group
- Developed an internal *end-to-end recommendation engine* to ease the data cleaning, preprocessing and model training pipelines

PRACTICAL SKILLS

Machine Learning
- Classical algorithms for supervised and unsupervised learning problems
- Probabilistic Inference
- Matrix Factorization
- Gaussian Mixture Models (GMM)
- Normalizing Flows
- Robustness
- Autoregressive Models
- Markov Chains
- Hidden Markov Models (HMM)
- Temporal Point Processes (TPP)
- PageRank

Deep Learning
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory Networks (LSTM)
- Attention and Transformers, GPT
- (Variational) Autoencoders (AE & VAE)
- Generative Adversarial Networks (GAN)
- Graph Neural Networks (GNN)
- Transformers
- Large Language Models (LLMs)

Programming Languages
- Python
- R
- SQL
- Scala

General
- Linux
- MacOS
- Git
- Bash
- PyCharm

Libraries
- numpy
- pandas
- scipy
- pytorch
- chainlit
- langchain
- scikit-learn
- tensorflow
- keras
- pyspark
- mlflow
- nltk

ACADEMIC BACKGROUND

PhD in Applied Mathematics
ENSAE PARIS
OCTOBER 2024 - PRESENT

- **Thesis Topic**: "*Statistical Properties of Generative Modeling Algorithms under Structural Assumptions*" supervised by *Professor Arnak Dalalyan*
- Teaching Assistant for the courses "*Advanced Machine Learning*", "*Theoretical Machine Learning*" and "*Statistics 1*"
- Participated in seminars, gave a talk on "*Statistical Guarantees of Generative Models*"

MS in Mathematics in Data Science
TECHNICAL UNIVERSITY OF MUNICH (TUM)
OCTOBER 2019 - JUNE 2022

Master's Thesis: "*Unpaired Data Translation between Multiple Incomplete Domains*" (in cooperation with BMW) supervised by *Prof. Dr. Stephan Günnemann* and *Aleksei Kuvshinov*
Other Projects:
- "Deep Learning for Genetic Risk Prediction" (cooperation with Helmholtz Zentrum München) supervised by Dr. Matthias Heinig
- "Depth Completion with Uncertainty Estimation" supervised by Prof. Dr. Matthias Nießner and Barbara Rössle

BS in Computer Science
AMERICAN UNIVERSITY OF ARMENIA (AUA)
SEPTEMBER 2014 - MAY 2018

Capstone Project: "*Low-Rank Bayesian Matrix completion using Gaussian Hierarchical Prior*" supervised by *Prof. Dr. Arnak Dalalyan*

PUBLICATIONS

V. Arsenyan, **E. Vardanyan**, and A. Dalalyan, "*Assessing the Quality of Denoising Diffusion Models in Wasserstein Distance: Noisy Score and Optimal Bounds*". **NeurIPS**, 2025.

E. Vardanyan, A. Minasyan, S. Hunanyan, T. Galstyan, and A. Dalalyan, "*Guaranteed optimal generative modeling with maximum deviation from the empirical distribution*". **ICML**, 2024.

A. Kuvshinov, D. Knobloch, D. Külzer, **E. Vardanyan**, and S. Günnemann, "*Domain Reconstruction for UWB Car Key Localization Using Generative Adversarial Networks*". **AAAI Conference on Artificial Intelligence**, 2022.

CONFERENCES AND FORUMS

NeurIPS San Diego 2025
- Poster presentation on "Assessing the Quality of Denoising Diffusion Models in W2 Distance: Noisy Score and Optimal Bounds"

Workshop on Generative Models in Science and Machine Learning, Heidelberg University 2025
- Delivered a talk on "*Statistical Guarantees of Generative Models*"

ICML Vienna 2024
- Poster presentation on "*Guaranteed Optimal Generative Modeling with Maximum Deviation from the Empirical Distribution*"

Armenian-German Research Summit on Intelligent & Data-Driven Systems Yerevan 2022
- Delivered a talk on "*Statistical Guarantees of Generative Models*"

DataFest Yerevan 2022
- Delivered a talk on "*Robustness in Deep Learning*"

CODASSCA Yerevan 2022
- Delivered a talk on "*Smart Car Key Localization using GANs*"

Barcamp Yerevan 2019
- Delivered a talk on "*Bayesian Methods for Neural Networks*"

DevFest Armenia 2018
- Delivered a talk on "*Optimization algorithms used in Neural Nets*"